

HEALTHCARE PRIOR AUTHORIZATION & INTEROPERABILITY PLATFORM

*Digitizing Prior Authorization Using FHIR-Based APIs
and HL7 Data Integration*

BUSINESS ANALYST PORTFOLIO PROJECT • PROJECT CHARTER

1. Project Background

MediConnect Health is a fictional healthcare technology company that aims to improve digital communication and information exchange between healthcare providers and insurance payers.

Currently, the prior authorization process may involve manual and disconnected workflows. For example, when a healthcare provider requests an authorization for a service such as an MRI, the provider must collect relevant patient, insurance, and clinical information and submit it to the payer for review. The payer reviews the request and communicates an authorization decision back to the healthcare provider.

Manual submission and communication processes can result in delays, incomplete information, duplicate data entry, and limited visibility into the authorization status. To address these challenges, MediConnect Health proposes a Healthcare Prior Authorization & Interoperability Platform that supports standardized digital information exchange using healthcare interoperability standards and APIs, including HL7 data integration and FHIR-based APIs.

2. Business Problem Statement

The current Prior Authorization process relies heavily on manual and disconnected workflows. This can result in delays in authorization processing, incomplete or missing information, duplicate data entry, and limited visibility into the status of authorization requests. These challenges affect healthcare providers, payers, and patients and can lead to inefficient communication and delays in the overall authorization process.

3. Business Objectives

- **Reduce Manual Work:** Automate and streamline prior authorization workflows to minimize manual intervention and improve operational efficiency.
- **Improve Turnaround Time:** Reduce the time required to process prior authorization requests and communicate decisions to healthcare providers.
- **Improve Information Completeness:** Ensure that required patient, insurance, and clinical information is complete and accurate before an authorization request is submitted.
- **Enable Real-Time Status Visibility:** Provide relevant stakeholders with timely visibility into the status of prior authorization requests throughout the authorization lifecycle.

- **Reduce Duplicate Data Entry:** Improve information exchange between connected systems to reduce redundant data entry and maintain data consistency.
- **Improve Provider–Payer Communication:** Support more efficient and standardized communication between healthcare providers and insurance payers.

4. Project Scope

4.1 In Scope

The following features and capabilities are included within the scope of the Healthcare Prior Authorization & Interoperability Platform:

- Prior Authorization request submission and management.
- Patient information required for Prior Authorization requests.
- Insurance and coverage information.
- Clinical information and supporting documentation.
- Authorization status tracking.
- Provider–payer data exchange.
- HL7 and FHIR-based healthcare data integration.
- API requirements for digital information exchange.
- Prior Authorization dashboard and operational reporting.

4.2 Out of Scope

- Actual insurance claims and payment processing.
- Real patient or personally identifiable healthcare data.
- Development of a complete Electronic Health Record (EHR) system.
- Real production deployment of the platform.
- Development of a mobile application.
- Clinical diagnosis or automated medical decision-making.

5. Key Stakeholders

The diagram below illustrates the end-to-end stakeholder flow, from the patient's initial need through to the final authorization decision.



6. Success Metrics

The success of the Healthcare Prior Authorization & Interoperability Platform will be measured using the following key performance indicators (KPIs):

KPI	Definition
Average Prior Authorization Turnaround Time (TAT)	Measures the average time taken from submission of a prior authorization request to the final decision.
Digital Submission Rate	Measures the percentage of prior authorization requests submitted through the digital platform.

KPI	Definition
Incomplete Request Rate	Measures the percentage of requests that require additional information because required data or documentation is missing.
Duplicate Request Rate	Measures the number or percentage of duplicate prior authorization requests submitted.
SLA Breach Rate	Measures the percentage of prior authorization requests that are not processed within the defined service-level agreement (SLA) timeframe.
Authorization Decision Rate	Tracks the percentage of requests resulting in Approved, Denied, Pending, or Additional Information Required statuses.
Average Processing Time	Measures the time spent processing requests at different stages of the prior authorization workflow.